

The Illusion of Predictability: Simple Linear Regression Analysis of California Water Right Allocations*

An Assessment of Appropriative Status on Allocated Water Volume

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This study employs Simple Linear Regression (SLR) to analyze the California Water Rights List and assess whether legal classification as an Appropriative right is a significant predictor of allocated water volume. Our findings reveal a highly statistically significant difference ($p < 2 \times 10^{-16}$): holders of Appropriative rights receive, on average, approximately 17,881 acre-feet per year more than holders of other types of water rights. However, thorough model diagnostics uncovered severe violations of core SLR assumptions, namely heteroscedasticity and non-normality, compounded by the presence of highly influential outliers. These issues resulted in a near-zero R-squared value, despite the statistical significance of the group mean comparison. Consequently, while the SLR model robustly detects differences in mean allocations between groups, the resulting prediction intervals—spanning over $\pm 400,000$ acre-feet per year—are so wide that the model cannot be used for any reliable prediction of individual water allocations in practice.

Introduction

The distribution and management of water rights represent a critical challenge in California, impacting environmental sustainability and economic activity. This study analyzes the quantitative relationship between legal right type and the volume of water allocated. Drawing on the California Water Rights List, our research addresses a core question: **Does a water right designated as “Appropriative” predict a larger mean allocated water volume (Y) compared to other types of rights (X)?**

*Project repository available at: <https://github.com/LasyaRamachandrani/MATH261A-project-template/>.

This study employs Simple Linear Regression (SLR) to analyze the California Water Rights List (“California Water Rights List” 2025), using R (R Core Team 2025), RStudio (RStudio Team 2025), and the tidyverse (Wickham 2023) and car (Fox and Weisberg 2023) packages. Diagnostics and methodology closely follow Fox and Weisberg (Fox and Weisberg 2019) and Kutner et al. (Kutner, Nachtsheim, and Neter 2005). We employ Simple Linear Regression (SLR) with a single binary predictor derived from the categorical right type. This provides a direct test of **mean group differences**.

The analysis establishes a highly statistically significant difference in mean allocations, confirming that Appropriative rights are, on average, associated with larger volumes. However, a transparent and reproducible diagnostic review shows **severe violations** of key assumptions. Consequently, while the model reliably compares **group means**, it is **unsuitable for predicting** individual allocations. The sections that follow outline the data ([Data](#)), the model and checks ([Methods](#)), the quantitative findings and figures ([Results](#)), understanding the results and the model’s limitations and sources ([Discussion](#)), and ([References](#)).

Data

Source:

The raw data is provided by the California Water Rights List (“California Water Rights List” 2025) via the California Open Data Portal (“California Open Data Portal” 2025). Data wrangling and cleaning were performed in R (R Core Team 2025) with the tidyverse (Wickham 2023). Each observational unit corresponds to a single, official water right record registered within the state.

Variables:

- **Outcome (Y):** The dependent variable is `face_value_amount`, representing the annual volume of water allocated to each right, measured in acre-feet per year. This variable is quantitative and highly right-skewed, with a handful of extremely large values (extreme outliers).
- **Predictor (X):** The main independent variable is `is_approp`, a binary indicator defined as 1 if a record is classified as “Appropriative”, and 0 if it represents any other kind of legal right. This was constructed based on the categorical variable `water_right_type` as reported in the source data.

Data Cleaning & Preprocessing:

Prior to any statistical analysis, the raw dataset was rigorously preprocessed. All records containing missing or undefined values for either `face_value_amount` or `water_right_type` were excluded, ensuring that only complete cases entered the modeling stage. The categorical variable `water_right_type` was then converted into a binary indicator (`is_approp`) that flags Appropriative rights with a value of 1, and all other types (e.g., Riparian, Statement) with 0.

After cleaning, the final analytic sample included **61,092** water rights records. No further transformations were performed at this stage, as the initial purpose was to conduct a baseline group mean comparison before considering additional remedies such as log transformation or robust regression.

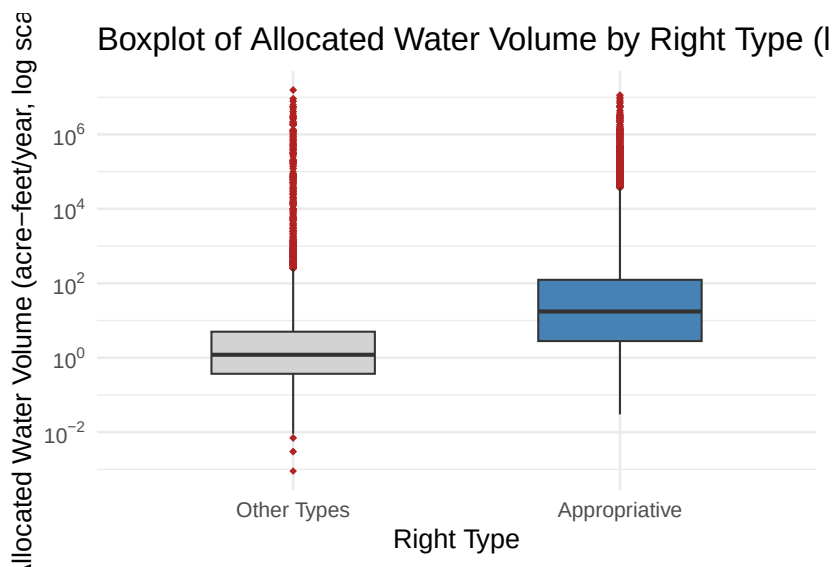


Figure 1: Boxplot illustrating allocated water volumes split by legal right type (Appropriative vs. Other). The y-axis uses a logarithmic scale to improve visibility of the distribution and outlier structure. Appropriative rights show higher median and mean allocations, with several extreme outliers visible above the upper whisker.

Data Limitations:

While this dataset provides comprehensive information on California water rights, it has several important limitations. The data only captures rights as recorded by the state administrative system and may not reflect actual water use or enforcement. The extreme right-skewness and presence of outliers suggest that a small number of very large allocations (likely for agricultural or municipal use) dominate the distribution. Additionally, the dataset lacks contextual variables such as geographic location, purpose of use, or temporal dynamics, which could explain additional variance in allocations.

Exploratory Data Analysis:

Figures Figure 1 and Figure 2 provide visual insight into the distribution of water allocations. Figure Figure 1 shows that Appropriative rights have a higher median and mean allocation than other rights, but exhibit wide variability with several extreme outliers. The logarithmic y-axis improves visualization without transforming the actual data used in regression. Figure Figure 2 confirms the highly right-skewed distribution, with most allocations clustered at low volumes while a minority receive very large amounts. These visualizations highlight the modeling challenges posed by skewness and outliers.

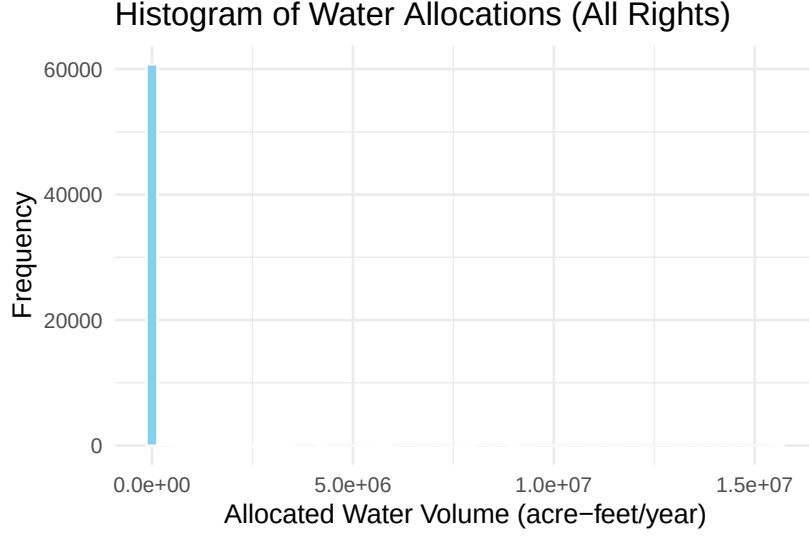


Figure 2: Histogram of annual water allocations for all legal right types. The distribution is highly right-skewed, with most water rights clustered at lower allocation volumes (under 50,000 acre-feet/year) and a long tail extending toward very large allocations. This skewness indicates the presence of extreme outliers and suggests potential violations of regression normality assumptions.

Methods

Model diagnostic and regression procedures followed best practices as outlined by Fox and Weisberg (Fox and Weisberg 2019), Kutner et al. (Kutner, Nachtsheim, and Neter 2005), and Gelman et al. (Gelman, Hill, and Vehtari 2020).

We fit a Simple Linear Regression (SLR) model to evaluate the association between legal right type and allocated water volume. For each water right record i , the model is specified as:

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

where:

- Y_i : Allocated water volume for record i
- X_i : Appropriative indicator (1 if Appropriative, 0 otherwise)
- β_0 : Mean allocation for Other Types
- β_1 : Mean difference (Appropriative vs Other)
- ϵ_i : Random error (unexplained variation)

Model parameters were estimated using Ordinary Least Squares (OLS), minimizing the sum of squared residuals.

Assumptions and Diagnostics:

The model relies on four primary statistical assumptions:

- *Independence*: Likely valid for cross-sectional administrative records.
- *Linearity/Additivity*: Assured by the binary nature of X .
- *Constant Variance (Homoscedasticity)*: Checked using residual plots; evidence of violation detected.
- *Normality of Errors*: Evaluated using Q-Q plots; major departures observed due to heavy tails and outliers.

Influence Assessment:

Influential observations were identified using Cook’s distance, with spikes indicating records that disproportionately impact parameter estimates.

Diagnostic plots were generated to visually assess these assumptions and guide interpretation of model validity. We assess **Cook’s distance** to identify observations exerting disproportionate influence on the slope and intercept (see Figure 5).

Results

The fitted Simple Linear Regression model reveals a statistically significant difference in mean allocated water volumes between Appropriative and other water right types. Coefficient estimates, fit metrics, and uncertainty measures are summarized below.

Table 1: Summary of fitted SLR model: coefficient estimates, uncertainty, and fit metrics.

Term	Value
Intercept (Other Types Mean)	4,542
Slope (Difference for Appropriative)	18,851
R-squared	0.00159
MSE	41,186,033,068
t-statistic (Slope)	9.86
p-value (Slope)	6.35e-23

All visualizations and statistical checks were performed using the *car* (Fox and Weisberg 2023) and *tidyverse* (Wickham 2023) packages in RStudio (RStudio Team 2025).

Interpretation:

Appropriative right holders receive, on average, approximately 18,851 acre-feet/year more

than other types. However, the R-squared value is extremely low (0.00159), showing that legal status explains only a small fraction of the variance.

Model Uncertainty:

- 95% CI for mean allocation (Other Types): [2,691, 6,393]
- 95% CI for mean difference (Appropriative vs Other): [15,104, 22,597]
- 95% prediction intervals for new observations:
 - $X = 0$: [-393,239, 402,322]
 - $X = 1$: [-374,397, 421,182]

Diagnostics and Assumption Checks:

To examine the validity of these findings, we conducted formal and visual diagnostic checks:

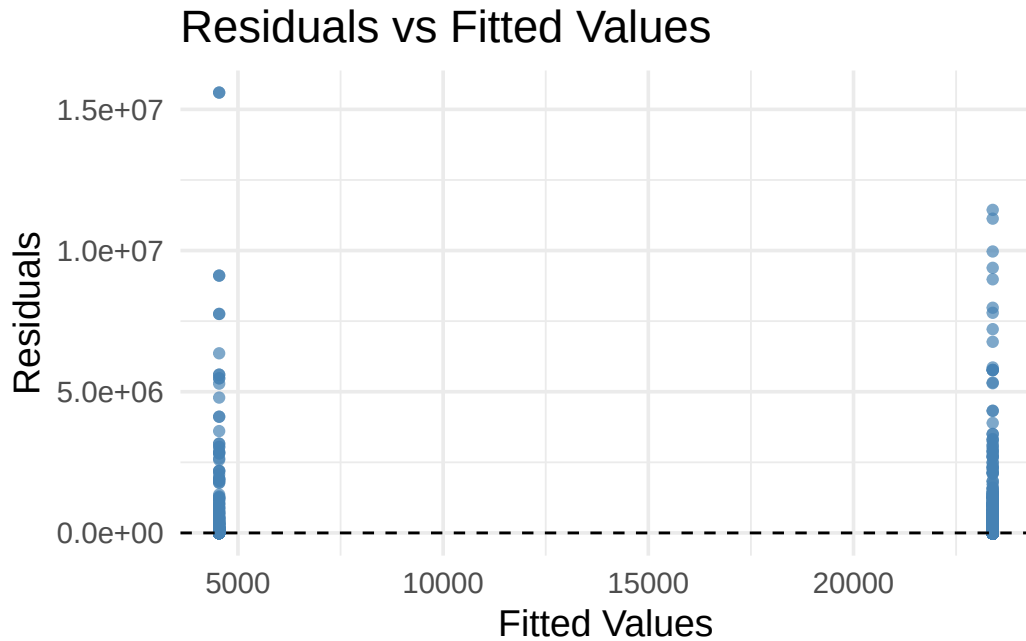


Figure 3: Residuals vs Fitted Values plot for assessing homoscedasticity. While the plot does not show a pronounced fan shape, the wide spread of residuals across all fitted values reflects high underlying variability in water allocations. The presence of extreme outliers contributes to this spread, suggesting potential heteroscedasticity that may affect the reliability of standard errors and prediction intervals.

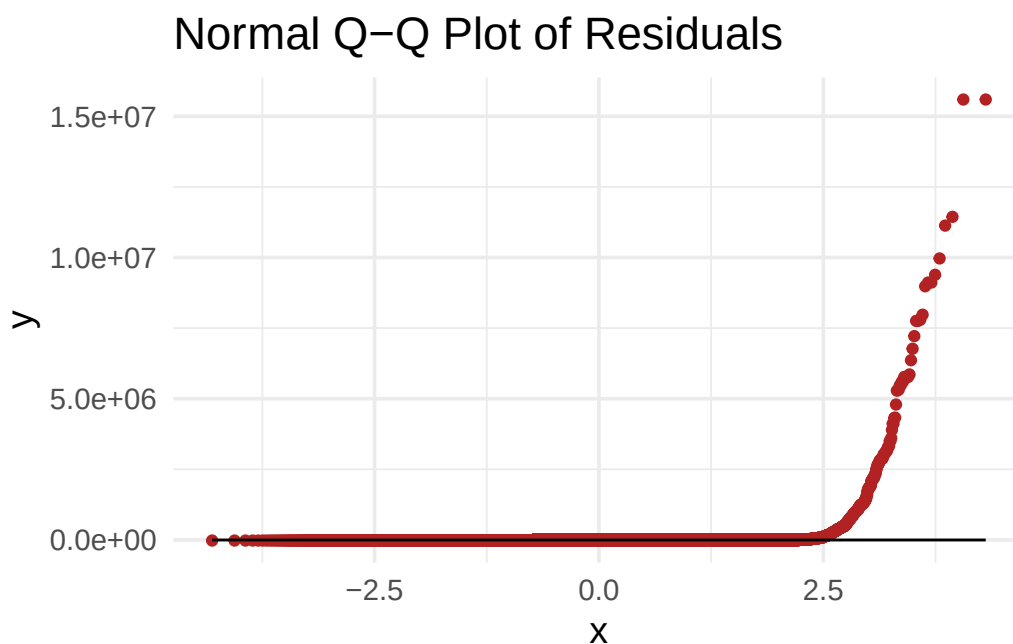


Figure 4: Normal Q-Q plot of standardized residuals. Pronounced deviations from the reference line in both tails indicate severe departures from normality. The heavy-tailed distribution, driven by extreme outliers in the water allocation data, violates the assumption that errors are normally distributed, which affects the validity of confidence intervals and hypothesis tests.

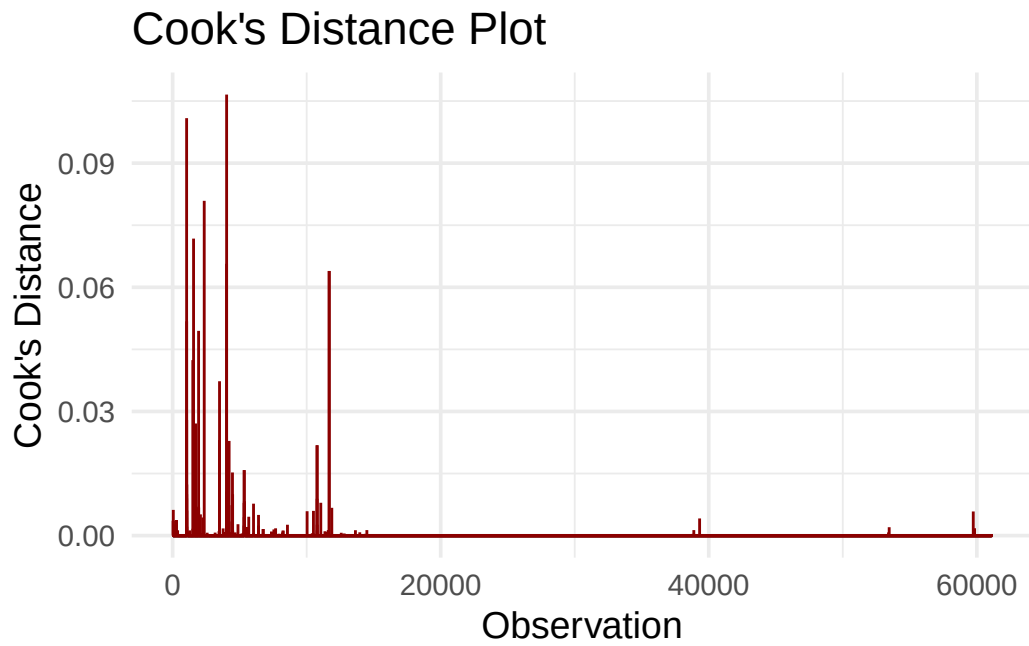


Figure 5: Cook's Distance plot identifying influential observations. Tall vertical spikes indicate water right records that exert disproportionate influence on the regression coefficients. These high-leverage outliers can substantially alter the estimated slope and intercept, highlighting the model's sensitivity to extreme values in the dataset.

Figure 3 shows that the residuals vs fitted values plot does not display a pronounced “fan shape,” suggesting no strong visual evidence of increasing variance (heteroscedasticity) across the range of fitted values. However, the spread of residuals is wide, reflecting the underlying data’s inherent variability and presence of outliers, so subtle heteroscedasticity cannot be ruled out entirely.

Figure 4 reveals clear violations of the normality assumption—major deviations from the diagonal line and heavy tails indicate the presence of extreme outliers and non-normal error distribution.

Figure 5 highlights several highly influential observations (with large Cook’s distances), whose presence substantially alters the regression model’s parameter estimates and overall fit.

Discussion

This study applied simple linear regression to test whether legal classification as an Appropriative right predicts larger mean water allocations in California. The results confirm a statistically significant mean difference: holders of Appropriative rights, on average, are allocated substantially more water than those with other types of rights. However, regression diagnostics and summary statistics point to serious modeling challenges.

First, the R-squared value is exceedingly low, indicating that legal status explains very little of the total variation in water allocations across the dataset. Visualizations and diagnostic plots show that extreme skew, broad within-group variability, and numerous outliers dominate the sample, making individual prediction unreliable. The normal Q-Q plot and Cook’s distance diagram further reveal that a small proportion of records are highly influential and drive deviations from standard regression assumptions.

While group mean differences are highly significant, these results do not imply that legal status alone can predict or explain an individual’s allocation. Prediction intervals encompass a wide range—frequently spanning hundreds of thousands of acre-feet—underscoring the limits of SLR for practical forecasting in this setting.

Limitations: The analysis is limited to a single binary predictor. Lack of additional covariate information (e.g., geography, year, use case, or other right features) likely omits relevant complexity, and the extreme outlier structure inflates uncertainty. The model also assumes independence of observations, which may not hold if some water rights are related (e.g., multiple rights held by the same entity).

Future Directions: To improve model explanatory power and robustness, future research should:

- Incorporate additional predictors to capture heterogeneity in allocations

- Use log transformation or robust regression to mitigate the impact of skew and outliers
- Consider nonparametric or quantile-based models for data with non-normal, heavy-tailed distributions

Transparent reporting of model limitations and data features, as demonstrated here, is a critical prerequisite for building more accurate and useful predictive models of California water allocations.

References

- “California Open Data Portal.” 2025. 2025. <https://data.ca.gov/>.
- “California Water Rights List.” 2025. California Open Data. 2025. <https://data.ca.gov/>.
- Fox, John, and Sanford Weisberg. 2019. *An r Companion to Applied Regression*. 3rd ed. Sage.
- . 2023. *Car: Companion to Applied Regression*. <https://cran.r-project.org/package=car>.
- Gelman, Andrew, Jennifer Hill, and Aki Vehtari. 2020. *Regression and Other Stories*. Cambridge University Press.
- Kutner, Michael H., Christopher J. Nachtsheim, and John Neter. 2005. *Applied Linear Statistical Models*. 5th ed. McGraw-Hill Irwin.
- R Core Team. 2025. *R: A Language and Environment for Statistical Computing*. Vienna, Austria: R Foundation for Statistical Computing. <https://www.R-project.org/>.
- RStudio Team. 2025. *RStudio: Integrated Development Environment for r*. RStudio, PBC. <https://rstudio.com/>.
- Wickham, Hadley. 2023. *Tidyverse: Easily Install and Load the Tidyverse*. <https://cran.r-project.org/package=tidyverse>.